Sample and preparation:

- Protein: Bovine serum albumin (BSA), ≥98% purity (e.g., Sigma A7906 or equivalent)
- Labeling: BSA-FITC or BSA–Alexa Fluor 488 at a low tracer ratio - approximately 1:100 to 1:200 (labeled:unlabeled by mass). The bulk of the BSA should be unlabeled. The fluorescent BSA serves as a tracer for self-diffusion in the crowded environment. I would prefer your expert lab handle labeling and sample preparation.
- Concentrations (6 samples):
    - 50 mg/mL (Fickian control)
    - 100 mg/mL (Fickian control)
    - 200 mg/mL (test)
    - 250 mg/mL (test)
    - 300 mg/mL (test)
    - 350 mg/mL (test, if soluble; BSA should remain stable to ~400 mg/mL)
- Number of replicates: At least 3 independent FRAP measurements per concentration (more is better for statistics)

Buffer and conditions:

- Buffer: PBS (phosphate-buffered saline, pH 7.4) or 150 mM Hepes, pH 7.4. Low ionic strength is fine; the key requirement is that BSA remains monomeric and does not aggregate. Which is why I want more trained hands than my own.
- Temperature: 25°C (room temperature)
- Sample format: Samples should be fully dissolved and equilibrated before measurement. BSA solutions above 200 mg/mL are viscous but should remain clear and isotropic if freshly prepared.

FRAP acquisition settings:

Critical requirements (non-standard):

- Bleach depth: ≥80% in the ROI. Standard FRAP often targets 50% bleach - that is insufficient here.
    - The front-propagation analysis requires a deep, well-defined bleach to cleanly resolve the recovery boundary. Please confirm your system can achieve ≥80% bleach in a single pass at the concentrations listed.
- Slow acquisition window: Recovery at 300+ mg/mL may take minutes, not milliseconds.
    - The total acquisition per measurement should extend to at least 15–20 minutes. Standard FRAP protocols that stop at 30 seconds will miss the signal entirely at high concentration.
- Full TIFF image stacks required. I need the raw image series (every frame, full field of view), not just the software's built-in ROI intensity trace.
    - The front-propagation exponent α is computed from the spatial profile of fluorescence at each time point — that analysis cannot be done from an intensity-vs-time curve alone. Please confirm you can export full TIFF stacks for each acquisition.

Data analysis expectations:

This is the critical part. I need two types of analysis, not just the standard one:

1. Standard recovery curve analysis: Fit the normalized fluorescence recovery $F(t)$ in the bleach ROI to extract the effective diffusion coefficient D_{eff} and mobile fraction. This is routine.
2. Front-propagation analysis (the key measurement): Measure the spatial extent of the recovery front $R(t)$ as a function of time. Specifically:
   - At each time point, measure the radius (or half-width) at which the fluorescence profile has recovered to 50% of its final value
   - Plot $R(t)$ vs t on a log-log scale
   - Fit a power law: $R(t) = A \times t^{\alpha}$
   - Report the exponent α for each concentration

At low concentrations (50–100 mg/mL), the exponent should be $\alpha \approx 0.50$ (standard Fickian diffusion). At high concentrations (200–350 mg/mL), I am testing a theoretical prediction that the exponent drops to $\alpha \approx 0.16$ due to degenerate (concentration-dependent) diffusivity. The distinction between $\alpha = 0.50$ and $\alpha = 0.16$ is the primary scientific objective of this experiment.

Does your analysis pipeline support front-propagation analysis? If not, providing the raw image stacks (TIFF or similar) for each FRAP acquisition would allow me to perform this analysis myself.

Deliverables I need:

- D_{eff} and mobile fraction at each concentration (standard output)
- The front exponent α at each concentration (the key measurement)
- Raw fluorescence intensity profiles (radial average from bleach center) at each time point, OR the raw image stacks
- A brief methods description (microscope model, objective, laser power, pixel size) for publication

Context:

I've attached a summary figure showing the output of my project: the universal mobility law $D(c) = D_0(1 - c/c_{max})^{\beta}$ fitted across 10 chemically distinct systems (hard-sphere colloids, BSA, sucrose, glycerol, PMMA, casein, PEG, dextran, Ficoll, hemoglobin), all $R^2 > 0.986$.

The FRAP front exponent $\alpha = 1/(3\beta + 2)$ is a zero-free-parameter consequence of this same law. The experiment you would be running tests whether the front-propagation prediction holds for BSA at high concentration, where $\beta \approx 2.1$ gives $\alpha \approx 0.16$ vs the Fickian $\alpha = 0.50$.

This experiment tests a zero-free-parameter prediction from a theoretical framework for concentration-dependent diffusion. If the results are positive, I would acknowledge Creative Proteomics as the experimental collaborator in the resulting publication.